

Evoker: Narrative-based Facial Expression Game for Emotional Development of Adolescents

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Figure 1: Loading scene of *Evoker*. Two observers of the expression monopoly company on the left of the image are stealing the citizen's emotions on the right side of the image.

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ABSTRACT

Evoker is a narrative-based facial expression game. Due to the COVID-19 pandemic, adolescents should be wearing masks in their daily lives. However, wearing masks disturbs emotional interaction through facial expressions, which is a critical component in emotional development. Therefore, a negative impact on adolescent emotional development is predicted. To solve this problem, we design a narrative-based game *Evoker* that uses real-time facial expression recognition. In this game, players are asked to identify

an emotion from narrative contexts in missions, and make a facial expression appropriate for the context to clear the challenges. Our game provides an opportunity to practice reading emotional contexts and expressing appropriate emotions, which has a high potential for promoting emotional development of adolescents.

CCS CONCEPTS

• **Human-centered computing** → **Interaction techniques**; *Scenario-based design*; • **Applied computing** → Education.

KEYWORDS

facial expression recognition, narrative-based game, emotional interaction, adolescents emotional development, COVID-19

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1 INTRODUCTION

Humans have six fundamental emotions (happiness, sadness, anger, fear, disgust, and surprise), and each emotion is expressed through facial expressions [7]. In addition, facial expressions convey emotional contexts of conversations [12, 29] and reflect an individual's psychological state [18]. Therefore, facial expressions play a decisive social function in interpersonal communication and relationships.

However, after the outbreak of the COVID-19 pandemic, people always have to wear masks. As the face is covered by the face mask, it has become difficult to understand emotional contexts. Therefore, people should recognize emotions by catching clues other than facial expressions. In particular, this social change has a severe impact on adolescent emotional development. Due to social blocking, adolescents in this pandemic period do not have the opportunity to experience emotional interactions [3]. However, if proper emotional development is not achieved in adolescence, it is difficult to return to a normal level later [32].

We propose *Evoker* which can support healthy emotional interpretation and expression through game play. For the long term effect of our game, it may help emotional development during adolescence. In our game, players experience a story that includes various emotional situations, and they have to make an appropriate facial expression for each mission which delivers emotional contexts only with body gestures and audio effects related to the situations. Through this experience, players can learn how people feel in certain situations. In other words, our game has a transformative potential that it helps foster a lack of emotional diversity in adolescence due to social disconnection in the time of the COVID-19 pandemic.

2 RELATED WORK

2.1 Importance of Facial Expression

Facial expression is an important factor in social function and relationships because it conveys the semantic content as well as the

emotional context of the conversation [12, 18, 29]. As fundamental emotions of humans are expressed through facial expressions [7], facial expressions have a significant influence in various ways.

However, COVID-19 has completely changed interpersonal communication. It is mandatory to wear a mask, but it hinders the chance to see and understand one's facial expressions during conversations [26]. In particular, as adolescents were unable to experience enough social relationships and emotional contexts [3], the number of adolescents with psychiatric disorders increased [15]. Previous literature also suggests that adolescence is the appropriate period for emotional development, and it is very difficult to make up for it later [32]. Therefore, adolescents in the pandemic period are vulnerable in that their emotional development can be damaged seriously.

Fortunately, making facial expressions helps develop emotions. Emotions can be already generated internally and expressed externally through facial expressions and actions [38], but external actions like voluntarily making facial expressions also can determine emotions [23]. Based on this, we expect the educational effect that the players can understand their feelings by voluntarily making facial expressions, and further enhance their understanding of others.

2.2 Facial Expression Recognition

Due to the multipurpose usefulness of facial expressions in various fields, such as human-computer interaction, augmented reality, and virtual reality, many studies have been conducted on the facial expression recognition problem since the 1990s [21]. Human emotions can be divided into six basic emotions [7], and therefore the facial expression recognition problem can be solved as a classification problem.

In conventional approaches, facial expression recognition consists of face detection, facial component detection, feature extraction by geometric methods, and classification using traditional classifiers like optical flow [25]. However, recent deep learning technology using a convolutional neural network (CNN) architecture has enabled end-to-end learning which can directly recognize facial expressions from a given image in real-time [24, 36]. Today facial expression recognition can be done in a few milliseconds, and thus it can be used in applications that require real-time interaction like games.

2.3 Related Games

2.3.1 Facial Recognition in Games. Recently, there are research to improve and expand the range of controllers using facial interactions. Microsoft Xbox introduces *Xbox Adaptive Controller* for game players with physical disabilities [1]. Also, studies on hands-free video game controllers reveal that facial expressions can be used to embrace the physically disabled players [30, 31]. In *the Same Boat (2019)*, which is a cooperative game using facial interaction, proved that facial expressions play a bridging role in intimacy and faith between two players. *Menti's Journey (2021)* uses facial expressions as an innovative interface for children under 10 years old [10]. Finally, games that connect emotions to weather have an innovative point that positive expressions like happiness were used to grow flowers, and negative expressions were used to wither

Table 1: Emotions, Places, and Mission Descriptions for Main Missions

Emotions	Places	Mission Descriptions
Happiness	House	Friends watch the soccer game together on a dark night. A voice of "Goal" is heard from the TV broadcast. Friends are all cheering and rejoicing.
Sadness	Hospital	The machines make sounds in the hospital. On the sickbed, a seriously injured person is bleeding and lying down. Someone who seems like a caregiver next to him is sobbing.
Anger	Mall	When entering a complex market, two men are quarreling loudly. It seems like they are fighting to buy the last sale one.
Surprise	Pub	A pub where zombies appear.
Disgust	Bathroom	A man is vomiting in front of the toilet because of a hangover last night.

flowers [2, 22]. Our game is differentiated from existing games in that it provides players various emotional contexts and let them make facial expressions appropriate for the current context to play the game.

2.3.2 Narrative-based Games. Narrative-based games have the advantage of enhancing the fun that the users feel [20]. In addition, interactive narratives are useful to generate emotions in games [13, 16]. *Seoul 2033 (2018)* provides options allowing players to organized their own stories. Also, *Disconnected (2017)* is a short horror game that brutally visualizes the absence of communication to sarcastically emphasize the importance of communication. Similarly, our game enhances the fun that players feel by providing narratives related to emotions.

2.3.3 Games with Emotions. Emotions are widely used in game applications dealing with mental regulation and emotion education. Research results on the positive influence of games on cognitive states present that game factors, such as levels, rewards, and narrative, help improve mental health [5, 28, 35]. Also, studies show that games can have a positive effect on their emotional state by providing other space for emotional expression to adolescents [14] and even children with autism [19, 34].

However, to the best of our knowledge, there are only few games for the emotional development of adolescents. Emotional development in adolescence could be naturally achieved in social situations like schools before the pandemic, but it became difficult due to the social disconnection. Since we take seriously the emotional and developmental problem of adolescents, we present an emotional game for adolescents.

3 GAME DESIGN

Evoker's primary goal is to make the player understand the given emotional context and make appropriate facial expressions. To make facial expressions a game element, we use them as the main interaction means. To this end, we designed game concepts and applied those concepts to the implementation with facial expressions.

3.1 Game Concept

3.1.1 Game Scenario. To use emotions as the subject matter of the game, we first created a background story. We set the similar environment to adolescents experience in the ongoing COVID-19 pandemic. We established a fictional world in which emotions

are used as an energy source. While emotion monopoly company extorts citizens' emotions, player is the only person who remembers facial expressions in the world. At the same time, as an *Evoker* performs missions to recover the lost facial expressions of the world. In summary, we set up a concept that the player should carry out missions to tell people emotions while not being taken their emotions by the company.

Among the basic six emotions, fear is excluded in this game. First of all, the Facial Action Coding System describes the relationship between the facial expressions and facial movements [8], and fear and surprise share at most four action units [4]. Also, although fear needs to be developed because it is an important adaptive emotion that has a decisive impact on human survival [6, 27], it is not ethically appropriate to induce excessively fearful emotions toward adolescents when using games for educational purposes. Therefore, our game uses five emotions (happiness, sadness, anger, surprise, and disgust) excluding fear.

Based on the five emotions, the emotionless non-player characters (NPCs) express the situation through nonverbal cues like voices and gestures. Then the player interprets which emotion is suitable for the given signals and makes an appropriate facial expression. The emotions, locations, and descriptions of each are shown in Table 1.

Players watch cinematic videos which explains the story before and after the game (see Figure 2 for detailed scenes). Text is also used to deliver the storyline, but we expected that visualization with the characters and backgrounds could be more helpful to understand the story clearly. Also, we additionally edited the videos for emotional completeness so that players feel more immersed before starting the game.

3.1.2 Game Genre. Since the player moves to several locations to perform missions while avoiding surveillance by enemies, the stealth game genre was suitable to reflect our main concept. Therefore, it is necessary to plan moderate level designs, enemies, and various strategies to avoid enemies [33].

Due to the nature of the genre, it is important to complete the mission as quickly as possible while avoiding surveillance. However, rushing can cause players not to have enough time to read and understand emotional contexts. In other words, if players have to make a facial expression while being exposed to enemies, players will not be able to think carefully about the emotional context. This discomfort will not be helpful for emotional development, which



Figure 2: Scenes from cinematic game trailer video. Image (a)-(e) are disgust, sadness, happiness, surprise, and anger scene, respectively.

is our first purpose. To avoid this problem, we designed the main missions to be carried out in internal places without external threats. Also, the NPCs of the indoor missions have no facial expressions but they deliver the emotional contexts through other nonverbal cues like body gestures. Therefore, the player experiences interpreting the emotional context without facial expressions, and furthermore, players can practice expressing emotions with facial expressions. See the flow chart in Appendix A.

Still, to avoid observers as a fun element of a stealth game, players should remain in external sites while keeping the consistency of interactions through facial expressions. Therefore, we designed the enemies as observers of the company who are moving around the city to extort people's emotions. They usually move around the designated area, but when the player gets closer to them than a certain distance, they chase the player leaving the designated area. Also, when the player is much closer than that, observers force the player to express a specific emotion. At this time, the player must make an appropriate facial expression that the observer requires. If the player succeeds in making a proper facial expression within the limited time, the lives of the player decrease 1, otherwise 2.

Hintman characters help the player find the next mission site. They look ridiculous like clowns who always smile since they have lost their emotions while smiling. The player has to explore the map and find a specific place to perform the mission, and the objects that the Hintman hands over as compensation for his mission can be used to find the next mission location. Similar to observer design, the mission of the Hintman characters is also to make an appropriate facial expression for consistency of interactions.

3.1.3 Four Core Elements of Game. Finally, we designed four core elements of the game (challenge, theme, reward, and progress) [11] (see Table 2 for more details). At this point, the subject that covers over three concepts except the challenge is color. First of all, we set an aesthetic concept that represents a state without emotions as grayscale and a state with emotions as colorful colors. In other words, when the player succeeds in a mission, then the NPC recovers the emotion and this is visually shown through colors (see Figure 3). Naturally, this visualization serves as a reward that adds the visual richness of the game. At the same time, the visualization acts as a device of progress that shows which parts of the city are cleared and which are not.

Table 2: Four Core Elements of Our Game

Game Element	Game Features
Challenge	Players should clear the mission which is to teach facial expression to NPCs, while avoiding the observers.
Theme	Grayscale colors represent the emotionless state while colorful colors represent the emotionally recovered state.
Reward	Coloring the object around themselves when players succeed in a mission. If players clear all the missions, they can reach the final goal which is breaking down the emotion monopoly company and making the world colorful again. Also, players can obtain clues for the next mission from the Hintman.
Progress	Players can check their progress through the color of the world as well as the map, and the emotion gauge which shows the remaining lives of the player.

3.2 Game Implementation

Evoker was built with Unity¹, and it is compatible with Windows OS. We purchased all assets for the map, characters, and sound effects in Unity Asset Store², and we produced additional character animations, textures, and other materials by ourselves for a more diverse and clear conveyance of emotional contexts. Also, cinematic videos were filmed by directing a camera walk suitable for the overall mood using Unity Cinemachine³ and additionally edited for higher emotional completeness.

For facial expression recognition, we use the FER⁴ module implemented in Python. FER computes the statistical confidence scores over 7 emotions (6 basic emotions and neutrality) from a given face image such that each confidence score is in range between 0

¹<https://unity.com/>

²<https://assetstore.unity.com/>

³<https://unity.com/unity/features/editor/art-and-design/cinemachine>

⁴<https://pypi.org/project/fer/>



Figure 3: Game scenes of *Evoker*. If the player completes the mission, scene will changes grayscale(a) to rich colors(b).

and 1 and the sum of all confidence scores is 1. FER provides both a haar cascade classifier and a multi-task cascaded convolutional network (MTCNN) for face detection. The detection performance is better in MTCNN, but the computation amount is greater than that of the Haar cascade classifier. Therefore, we use the Haar cascade classifier because our game requires real-time facial interactions. Also, since it is impossible to run the FER module in C#, which is the language used in Unity, the probability values computed from the FER module were sent to the game engine by local communication using transmission control protocol (TCP).

As a criterion for a success of a mission, we use the average confidence score for the previous one second. In other words, facial expression recognition is performed every frame, and the average confidence score over 30 frames must be above a predetermined threshold value because the FER module has a running speed of 30 frames per second (FPS). The thresholds for each emotion were tentatively set as confidence scores suitable for smooth game progress as a result of several internal tests. Threshold values are shown in Table 3.

Table 3: Threshold Confidence Scores

Happiness	Sadness	Anger	Surprise	Disgust
0.5	0.38	0.31	0.3	0.15

4 PILOT TEST

The purpose of the pilot test is to obtain user experience data and set the facial expression threshold values of confidence scores for each emotion. This questionnaire pilot test was conducted twice in watching demonstration video as the first test, and actual play mode

as the second test. Each participant participated in the above experiment only once. Participants under the age of 18 were given with consent of their parents after sufficiently explaining the purpose and method of the test.

In the case of experiments by watching the gameplay video, we conducted the tests on adolescents to verify whether our game is effective for our target players. Four test subjects (one female; mean age 17.25; age range 18–21 years) participated in the first pilot test, and they responded to the survey after watching the gameplay video. As a result, the influence of the game on their emotional diversity averaged 3.5 points, and the fun averaged 3.75 points, which are quite positive. Also, we could get positive feedback in that facial expressions are used as a controller. However, it was pointed out that the overall user interfaces and directing are not natural but rather interfere with immersion. Therefore, we improved the user interface and the quality of both the intro and outro video to enhance the player’s understanding. As a result, the influence of the game on their emotional diversity averaged 3.5 points, and the fun averaged 3.25 points, which are quite positive (see the details in Table 4).

Table 4: Result of the Pilot Tests

Category	First Test (N = 4)		Second Test (N = 7)	
	Mean	SD	Mean	SD
Helpful for emotional expression	3.5	0.58	3.57	0.79
Fun	3.25	1.26	4.43	0.53
Graphic design	3.75	1.5	4.29	0.49
Replay intention	3.75	1.5	4.29	0.49

Like the first test, the second test was also conducted to observe whether our game is effective for adolescents. This time, unlike before, seven testers (four females; mean age 19.57; age range 18–21 years) played the game and completed a questionnaire.

Consequently, the influence of the game on their emotional diversity averaged 3.57 points (see the details in Table 4), so we could confirm that our game is helpful to achieve our goal, which is to help adolescents make natural facial expressions in various emotional contexts. Also, all the other categories scored higher than the previous test, so we could confirm that the gamification elements have developed a lot. Finally, we could get positive feedback about a unique means of interaction using facial expressions as before, and some players responded that our game could have a good influence in a modern society where people do not reveal emotions during the prolonged pandemic.

However, subjects answered that there were cases where our game could not recognize disgust even though they were doing it properly. To analyze the cause of this problem, the confidence scores of emotions required when the player is performing a mission were collected. As described in Section 3.2, the confidence score of a specific emotion required in a mission is obtained by each frame, and we use the mean and the standard deviation of the obtained values for each emotion. In Table 5, the *Mean* and *SD* columns show the mean and the standard deviation of confidence scores of each emotion. As a result, the average confidence score of disgust is

about 0.03, which is much lower than other emotions. Therefore, adjustment of threshold values remains necessary, and it would be better for balancing the difficulty level if they are personalized by measuring the player's facial expressions before starting the game.

Table 5: Analysis of the Confidence Scores of Emotions

Category	Mean	SD
Happiness	0.2119	0.3028
Sadness	0.1699	0.1394
Anger	0.1765	0.1561
Surprise	0.0975	0.1714
Disgust	0.0363	0.0645

5 CRITICAL REFLECTIONS

Our game has a transformative and transgressive potential in that it can help solve developmental problems in adolescents that have occurred in the context of the times. Although emotional education of adolescents is an important problem, it was a part that could be achieved through education in schools and social institutions, or through interaction within social peer groups. However, adolescents experienced social blocking due to the COVID-19, and their emotional development was significantly damaged due to the lack of prior research on the emotional education without social relations. Therefore, our game has a social value because it deals with the emotional education of individual adolescents, which is a point that has not been well dealt with in previous studies.

However, present omnibus structure has a limitation. Since each story is an independent event, players cannot experience a single continuous story. Therefore, if we modify the scenario that has connection between each scene, players can expect a greater sense of immersion. In addition, a consistent story can deal with mixed emotions. In fact, a person's feelings are not determined as a single emotion, and several emotions may be mixed [17, 37]. Also, multifaceted self is formed in adolescence [9]. On the other hand, the emotions we provide are limited to one emotion for each mission, so there may be limitations in our contribution to the development of adolescent emotions, and we expect it can be solved by providing an emotionally consistent story.

Since we are in the early process of development, our tests were a simple experiments in the form of a pilot test. As the game development progresses, we are going to conduct formal tests after the approval of the Institutional Review Board (IRB) as our game is developed further. Also, in order to bring out more entertainment to players, some extra polish will be added.

6 CONCLUSION

Facial expressions play an important role in social and emotional interactions. However, wearing mask during the COVID-19 pandemic disturbs adolescents properly understanding other's emotions and expressing their own emotions. This can lead to emotional development problems in adolescence.

To solve this social problem, we propose *Evoker* that can train emotional expression for adolescents through narrative-based progress

while utilizing the characteristics of the stealth game genre. In our game, the narrative includes several emotional contexts, and the players first have to interpret emotional contexts based on the given cues other than facial expressions. Then, the players are required to make a facial expression appropriate for the emotional context to clear the challenge. We have been able to confirm that our game can be effective in solving the social problems we perceive, and thus it is expected that our game has the potential to have a positive impact on emotional development.

As a novelty of our game, we present a new way of interaction through facial expressions. Furthermore, our game is differentiated from existing games that use facial expressions as a controller in that we provide emotional situations suitable for each expression so that the effect of emotional education can be enhanced. Finally, based on the psychological literature that voluntary facial expressions can arise emotions, it suggests a potential that might solve social problems that arise in the context of the times.

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A FLOW CHART

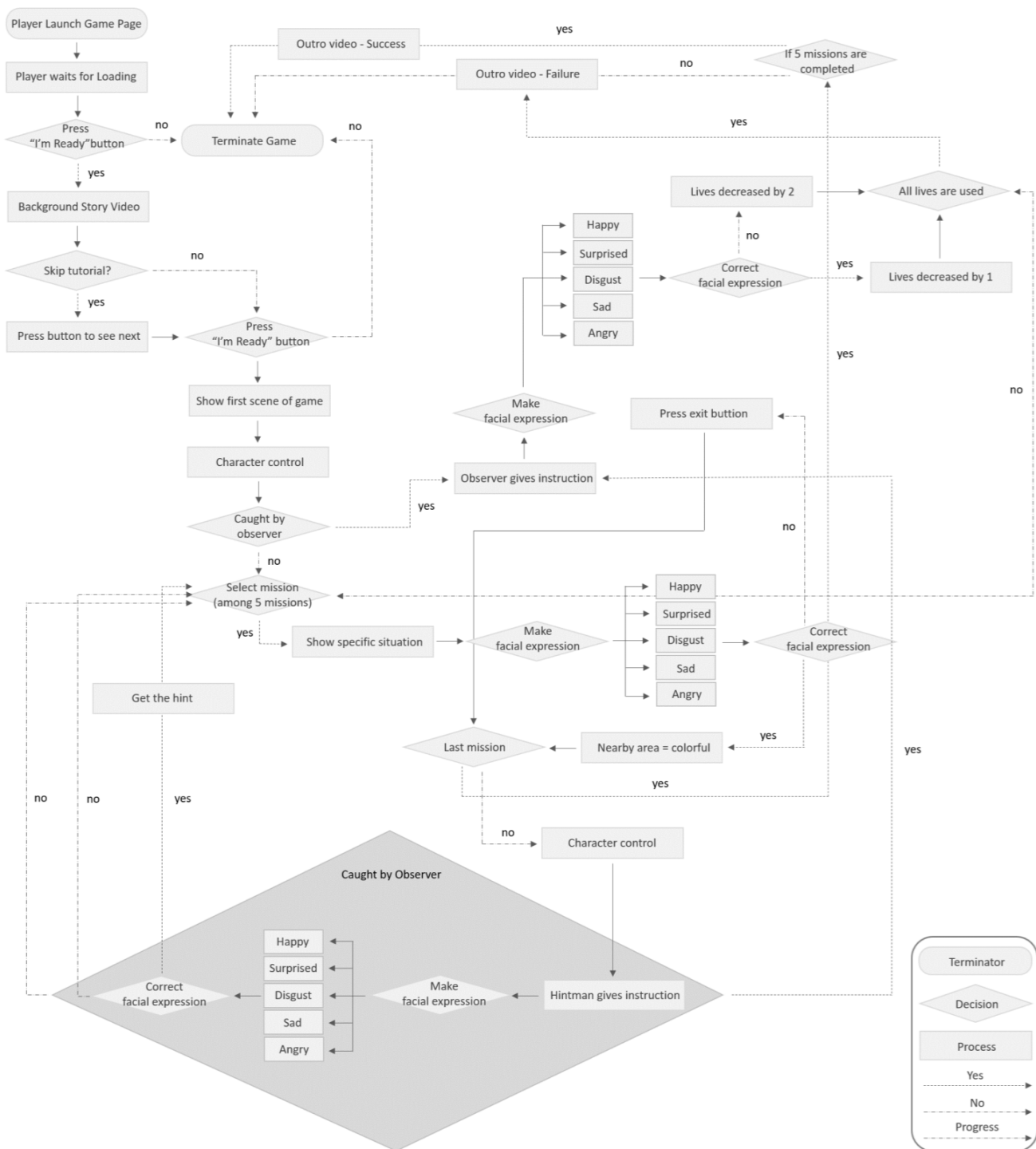


Figure 4: Detailed flow chart.